Borrow-Box

Software Requirements Specification

Table of Contents

Software Requirements Specification

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the functional and non-functional requirements of the **Online Renting System** developed using the Django framework. This document explains the system functionality, system constraints, user interactions, and interface requirements. It serves as a guideline for developers, stakeholders, and project evaluators to understand the system structure and expected behavior.

1.2 Scope

The Online Renting System is a web-based application that enables users to rent products from other users and also list their own products for rental.

The system promotes a **shared economy**, allowing individuals to generate income from unused items while providing affordable rental options to other users.

The platform provides features such as:

- User registration and authentication
- Listing items for rent
- Browsing and searching available items
- Rental request and booking management
- Payment options for renting items
- Admin approval and system management
- The system aims to create a convenient, transparent, and efficient rental platform.

1.3 Definitions, Acronyms and Abbreviations

Term	Description
Admin	System administrator responsible for managing the platform
User	A registered person who can rent or list items
Owner	A user who lists items for rent
Renter	A user who rents items
Django	Python-based web framework used for development

1.4 References

- Django Official
- Python Programming
- Tailwind css
- Razorpay Payment Gateway

1.5 Overview

This document provides a detailed overview of the Online Renting System including its overall description, system functionalities, requirements, and interfaces.

The document is structured into three major sections: Introduction, Overall Description, Specific Requirements.

2. Overall Description

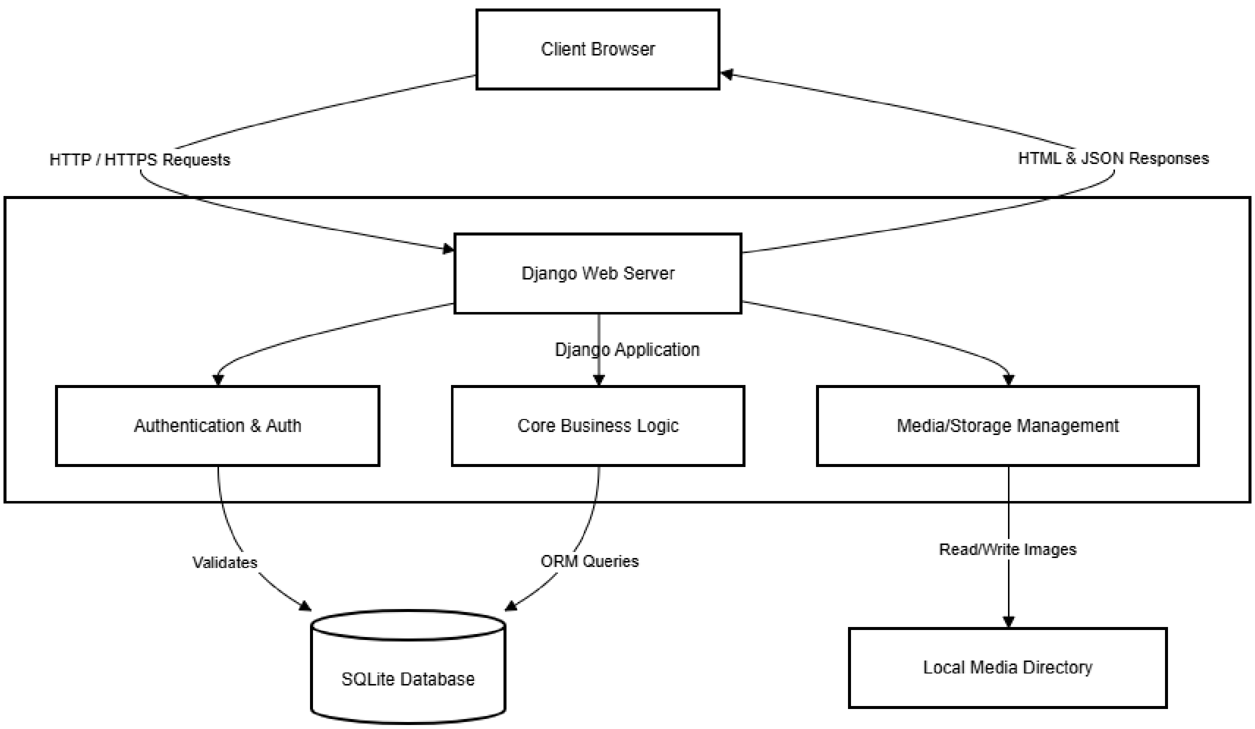
2.1. Product Perspective

The Online Renting System is designed as a **web-based platform** where users can rent items from others and list their own items for rental.

The system is built using the Django framework and follows the **Model-View-Controller (MVC)** architecture.

The system consists of the following main modules:
User Module
Item/Product Module
Rental/Booking Module
Payment Module
Admin Module

📌 System Architecture Diagram:



2.1.1. System Architecture Diagram

2.2. Product Functions

- Main functions of the system include:
- User registration and authentication
- Profile management
- Listing items for rent
- Browsing and searching items
- Sending rental requests
- Rental duration selection
- Booking status tracking
- Payment management
- Admin approval and system monitoring

2.3. User Classes and Characteristics

- **User**

  Can register and login
  Can list items for rent
  Can browse and rent items

- **Admin**

  Manages users and items
  Approves item listings
  Monitors rentals and payments

2.4. Operating Environment

The system operates in the following environment:
Operating System: Windows / Linux
Web Browser: Chrome, Firefox, Edge
Backend Technology: Python Django
Frontend: HTML, CSS, Tailwind css, JavaScript
Database: SQLite

2.5. Design and Implementation Constraints

The system must be developed using Django framework.
The application must run on web browsers.
The system must use a relational database (SQLite).

2.6. Assumptions and Dependencies

Users must have internet access.
Users must register before renting items.
Admin approval may be required before item listings become visible.

3. Specific Requirements

3.1. Functional Requirements

FR1 – User Registration

The system shall allow users to create an account by providing necessary details such as username, email, and password.

FR2 – User Login

The system shall allow registered users to login securely using their credentials.

FR3 – Profile Management

Users shall be able to update their personal profile information.

FR4 – Item Listing

Users shall be able to list items for rent with details including item name, category, description, rental price, availability period, and images.

FR5 – Item Browsing

Users shall be able to browse and search available items using categories and keywords.

FR6 – Rental Request

Users shall be able to select rental duration and send rental requests.

FR7 – Booking Status Tracking

Users shall be able to track rental status such as **pending, approved, rented, or returned**.

FR8 – Rental History

Users shall be able to view history of items rented or listed for rent.

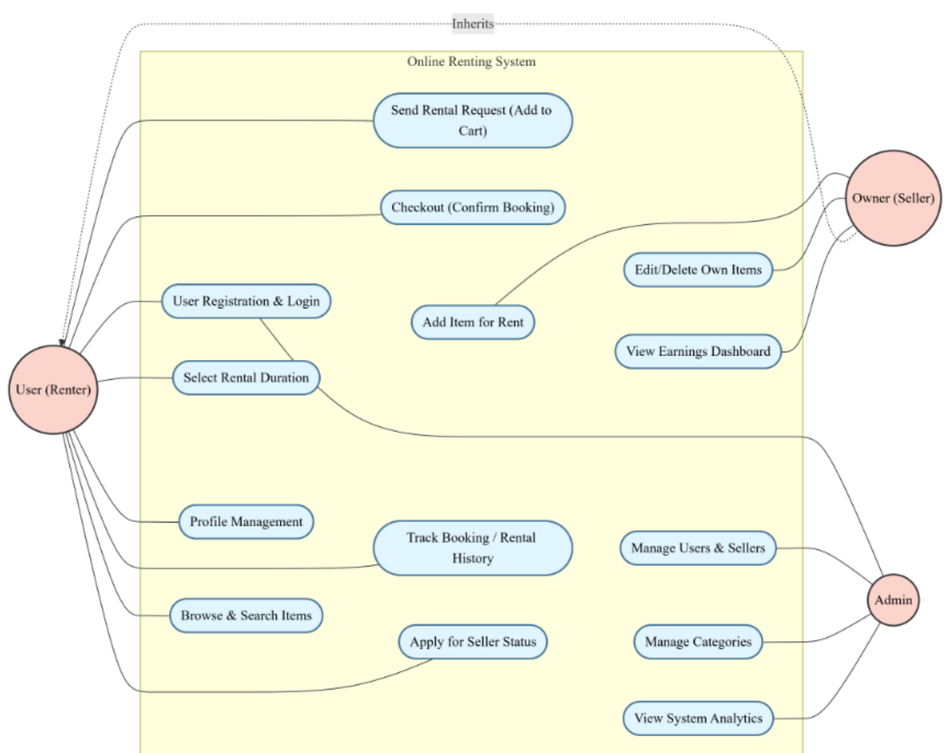
FR9 – Payment Management

The system shall support online payments or cash-based payment options.

FR10 – Admin Management

The admin shall be able to manage users, categories, items, and rental records.

📌 **Insert Use Case Diagram here**



3.1.1.1 Use Case Diagram

3.2. Usability Requirements
The system must provide a user-friendly interface that allows users to easily navigate and perform actions without technical knowledge.

3.3. Reliability Requirements
The system must maintain consistent performance and ensure secure storage of user data.

3.4. Performance Requirements
The system should respond to user requests within **2–3 seconds** under normal usage conditions.

3.5. Supportability Requirements
The system should be designed using modular architecture to allow easy maintenance and future enhancements.

3.6. Design Constraints
Django framework must be used for backend development.
System must support modern web browsers

3.7. Online User Documentation and Help System Requirements
The system should provide basic guidance and instructions to help users understand how to use the platform.

3.8. Interfaces
3.8.1. User Interface
Login Page
Registration Page
User Dashboard
Item Listing Page
Booking Page
Admin Dashboard
3.8.2. Hardware Interface
Laptop / Desktop
Smartphone

3.8.3. Software Interface
Django Framework
SQLite Database
Web Browsers

3.8.4. Communication Interface
HTTP / HTTPS internet communication